

Phylogenetic Analysis of Community Assembly of Fish Species in the Great Lakes

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Assignment 4 - BINF 6210: Software Tools
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The Laurentian Great Lakes provide ecosystem services that enhance quality of life

The Great Lakes ecosystems and their species communities provide a range of often underestimated services, including cultural, recreational, economic, and protective benefits (Steinman et al., 2017).

Rising water temperatures as a result of climate change have led to changes in fish phenology and community composition in the Great Lakes over the last 27 years, underscoring the urgency for targeted monitoring and adaptive management strategies to conserve native biodiversity and maintain ecosystem services (Piczak et al., 2025).



[Michael Furtman](#)

Few studies connect evolutionary processes with ecological interactions that dictate modern community assembly



Backenstose et al. (2024) examined the diversification of *Coregonus* species in the Great Lakes, showing how habitat availability and predator-prey interactions shaped their evolutionary trajectories and how species respond to environmental change.

While this work highlights the importance of evolutionary history for understanding individual species dynamics, the role of phylogenetic relationships in shaping multi-species community assembly across the Great Lakes remains largely unexplored.

Few studies connect evolutionary processes with ecological interactions that dictate modern community assembly

By combining molecular phylogenies with presence-absence data and calculating the Nearest Taxon Index (NTI), this analysis uses COI-5P phylogenetic relationships to connect evolutionary history with present-day ecological interactions within Great Lakes Fish communities.

The general hypothesis being tested in this analysis is that fish assemblages across the Great Lakes will follow patterns consistent with other freshwater systems and display non-random phylogenetic structure within communities (Bower and Winemiller, 2021).

Mechanistically, there are two competing hypotheses being tested:

1. **Phylogenetic Overdispersion:** When competitive exclusion and niche differentiation dominate, closely related species are less likely to coexist. Assemblages will therefore be phylogenetically overdispersed, and NTI values will be negative compared to the null model.
2. **Environmental Filtering:** When resource stress or environmental uniformity dominates, species sharing conserved traits are more likely to coexist. Assemblages will be phylogenetically clustered, and NTI values will be positive relative to the null model.

Data from Great Lakes Water Life and BOLD were joined

Great Lakes Environmental Research (GLERL) Water Life Explorer

- Database containing presence/absence information for species across the Great Lakes and adjacent rivers - selected the fish classification group
- 178 records, 40 variables
- Interested in:
 - Species name
 - Lake presence (Huron, Ontario, Michigan, Erie, Superior)
- Accessed November 24, 2025

Barcode of Life Database (BOLD)

- Database containing DNA barcode data for many taxa across the kingdoms of life - selected *Phylum Chordata*
- 746,080 records, 80 variables
- Interested in:
 - Species name
 - Markercode (COI-5P)
 - Nucleotide sequences
- Accessed November 24, 2025

The bioinformatics pipeline leveraged multiple software tools for complex end-to-end analyses

Data Acquisition & Cleaning

- Compiled fish species occurrence and COI-5P sequence data.
- Standardized names, converted presence - absence data, and joined species data with nucleotide sequences.

R Packages: tidyverse, janitor, httr (BOLD API access)



Sequence Filtering and Preparation

- Filtered nucleotide sequences by length and quality.
- Selected representative sequences per species and exported them in FASTA format for phylogenetic analysis.

R Packages: tidyverse, stringr, phylotools



Phylogenetic Tree Construction

- Aligned COI-5P sequences (MUSCLE, MEGA) and built a maximum likelihood tree (IQ-TREE2) (details to follow)
- Imported and visualized the tree structure to confirm taxonomic representation.

R Packages: ape, ggtree
Software: MEGA, IQ-TREE2



Community Diversity Analysis

- Created a species-by-lake community matrix to calculate phylogenetic diversity (PD) and species richness (SR).
- Visualized lake-level diversity and relationships between PD and SR.

R packages: picante, tidyverse, ggplot2, patchwork, ggrepel



Phylogenetic Structure and NTI Evaluation

- Computed NTI using null model simulations to assess clustering vs. overdispersion in each lake.
- Interpreted z-scores to infer environmental filtering or competitive exclusion

R packages: picante, ape, ggplot2



External software tools MEGA 12 and IQ-TREE2 were used for phylogenetic tree construction



COI-5P nucleotide sequences were exported from R as a FASTA file and aligned using MUSCLE within MEGA software.

- Gap open penalty of 400.00
- Gap extension penalty of 0.00

Alignment was visually inspected for quality assurance but not manually trimmed, as sequences were already filtered between 500–800 bp in R.

- Alignment file exported from MEGA was saved as fish_sequences_aligned_MUSCLE.fas

The alignment was then used in a substitution model selection analysis, which identified that GTR would be best.



IQ-TREE 2

MEGA-aligned dataset was used as input for IQ-TREE2 using GTR model (identified by MEGA as best model).

- GTR - General Time Reversible model
- F - observed nucleotide frequencies used
- G4 - substitution rates vary among sites with four discrete rate categories

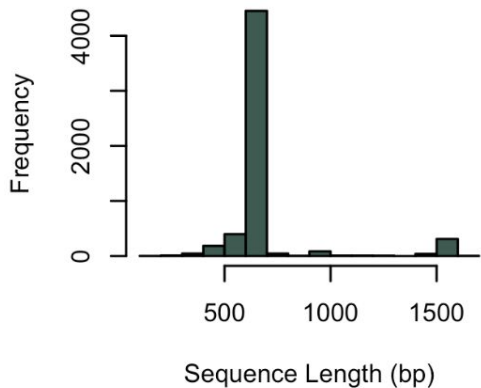
Phylogenetic reconstruction was then performed:

```
iqtree3 -s fish_sequences_aligned_MUSCLE.fas -m GTR+G  
-B 1000 -T AUTO
```

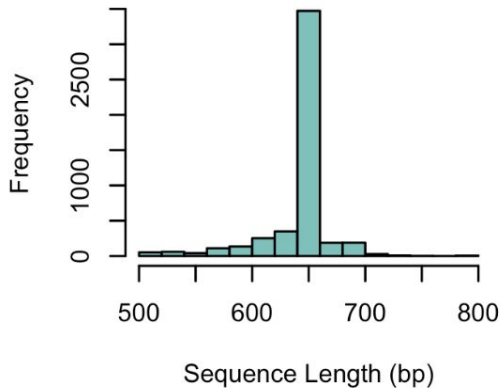
- Maximum likelihood (ML) analysis with 1,000 ultrafast bootstrap replicates
- Consensus tree (.contree file) and model report (.iqtree file) were used for downstream analyses

Figure 1. Progressive filtering yields a high-quality, uniform set of nucleotide sequences for the COI-5P locus

A: Before Filtering



B: After Sequence Length Filter



C: After Ambiguous Base and Gap Filter

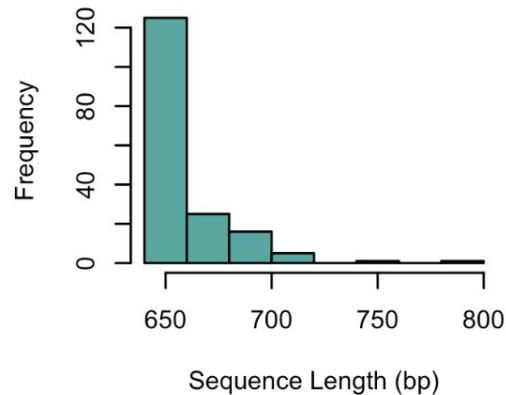


Figure 2. COI-5P-based maximum-likelihood phylogeny illustrates species-level relationships

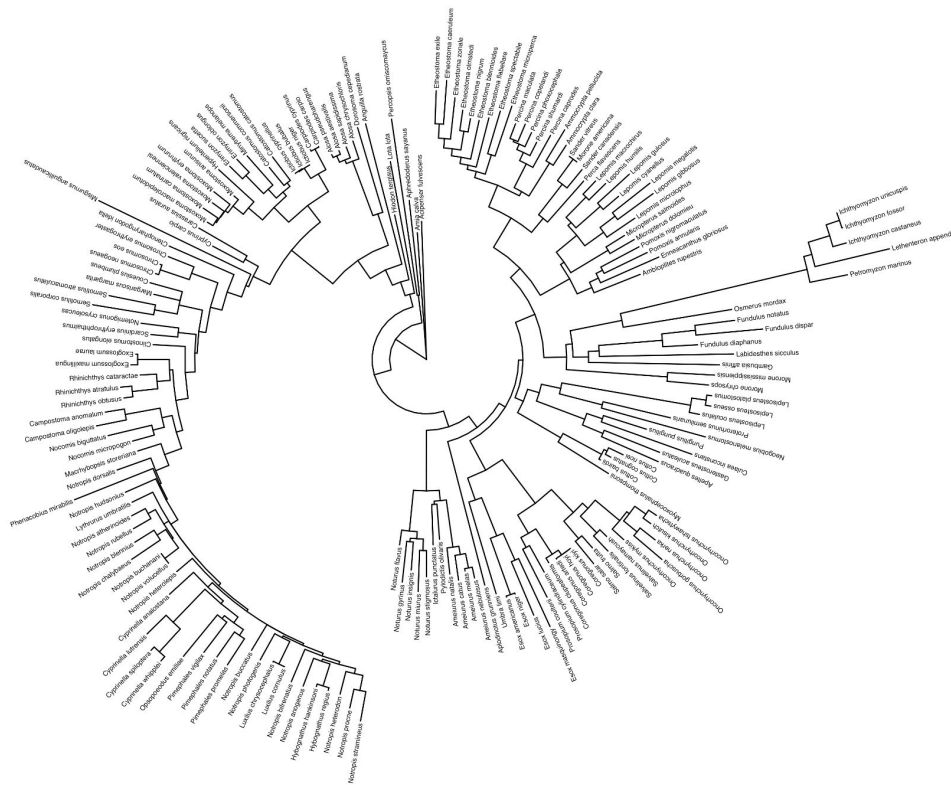


Figure 3. Community matrix highlights species overlap among Great Lakes with some lake-specific variation

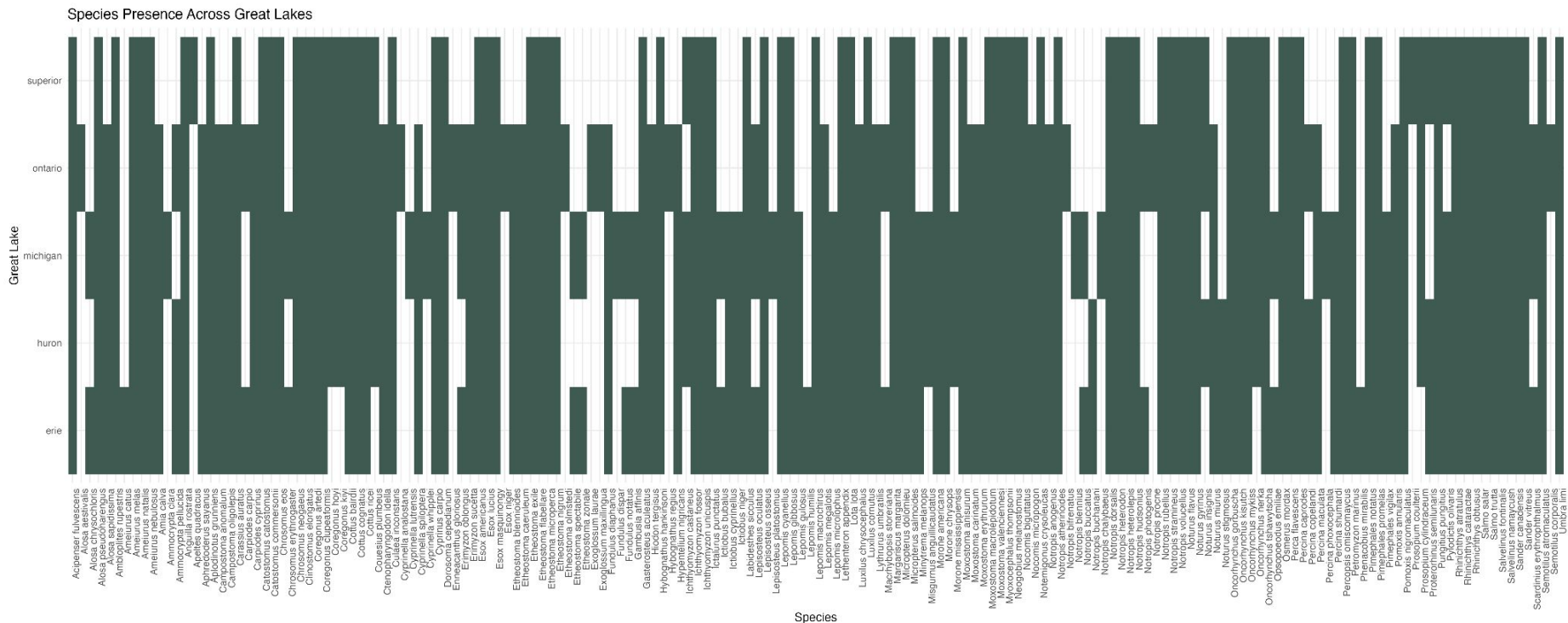
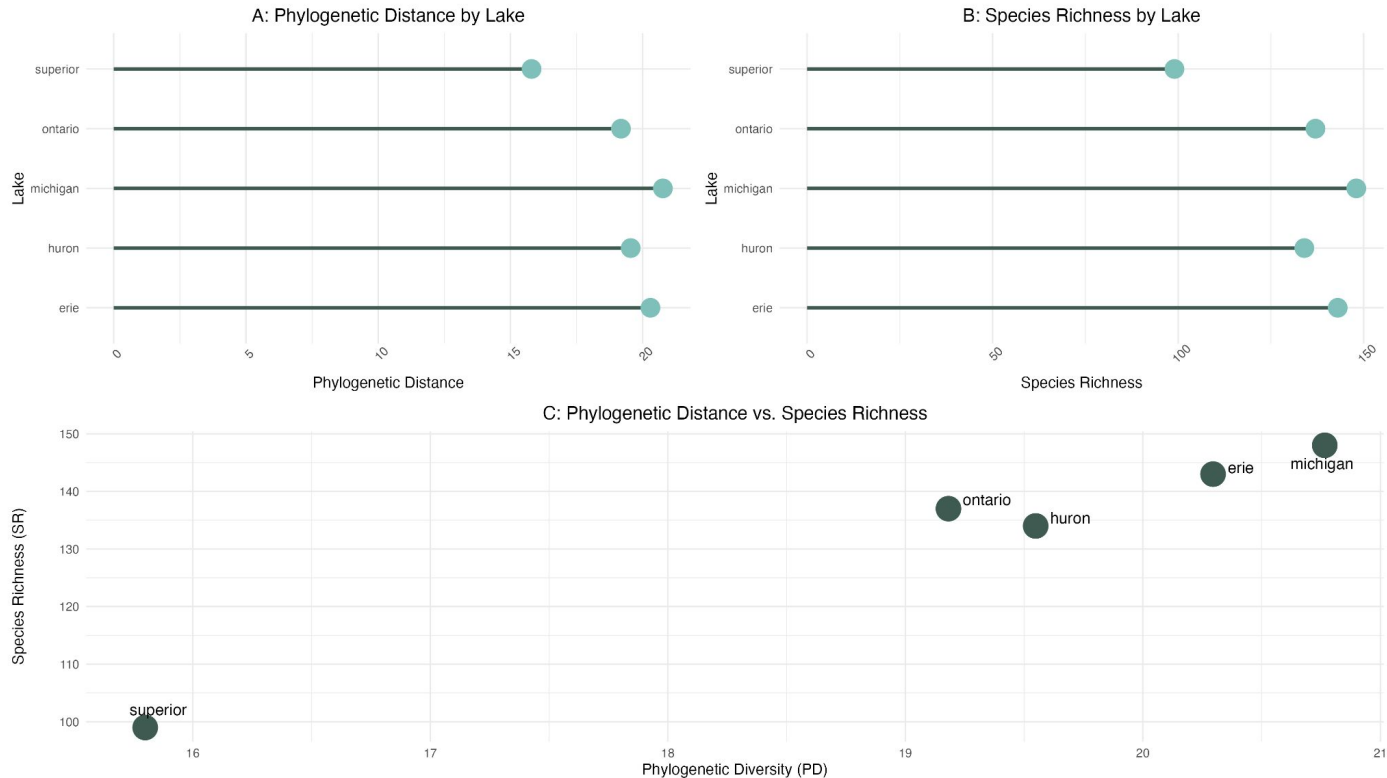


Figure 4. Great Lakes with higher fish species richness exhibit greater phylogenetic diversity



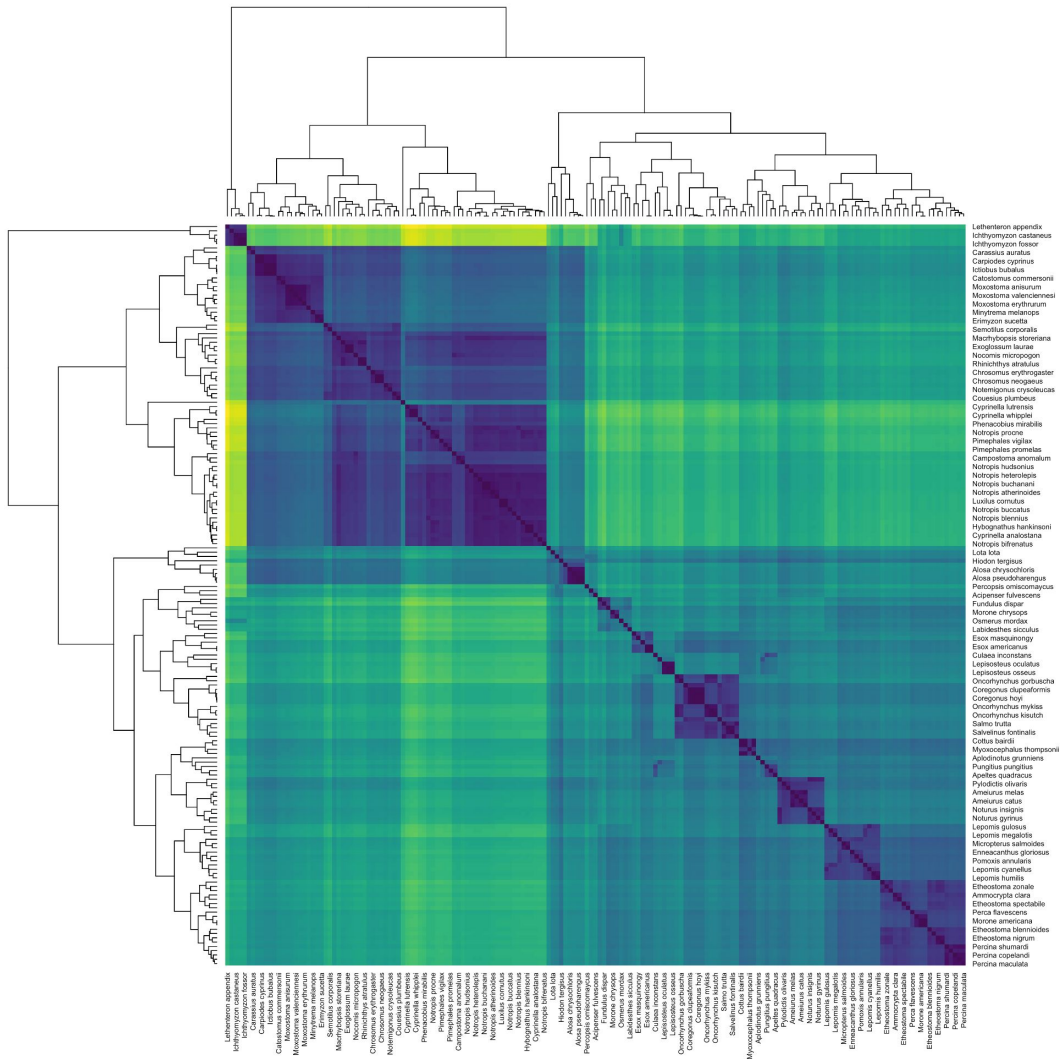
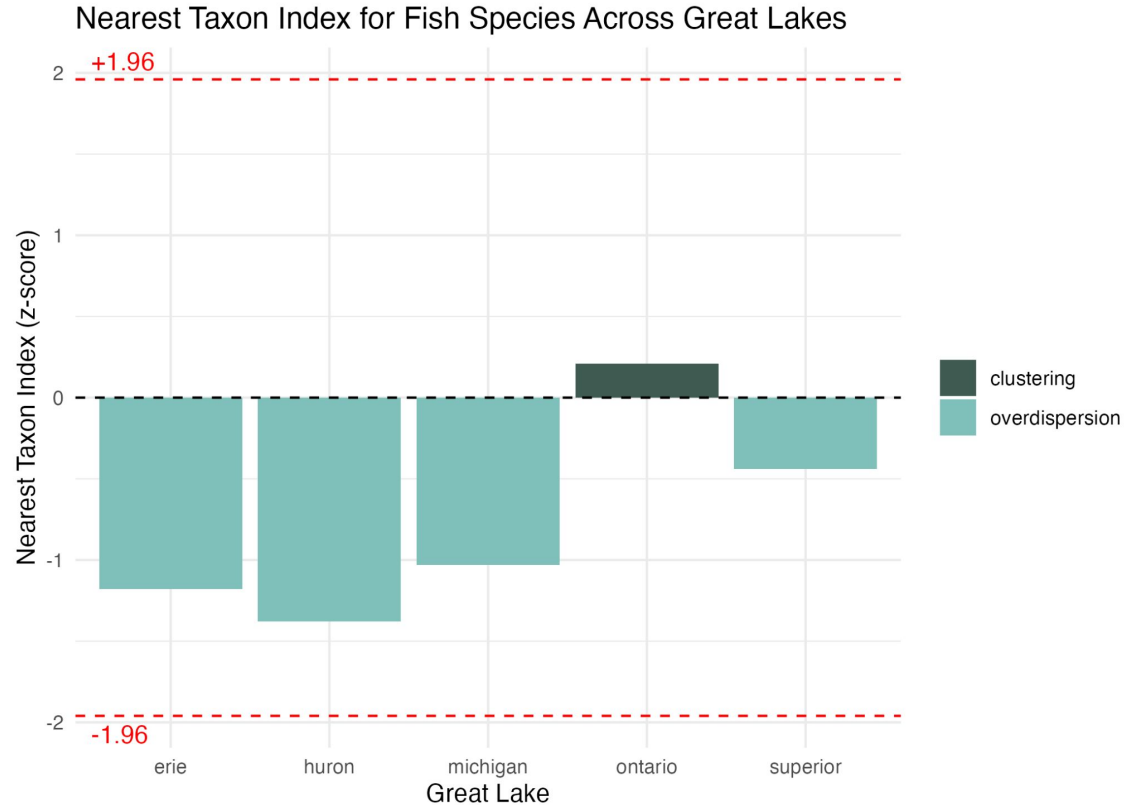


Figure 5. Phylogenetic distance matrix reveals clustering of related taxa and separation among major fish lineages

Figure 6. Nearest Taxon Index reveals phylogenetic overdispersion across most Great Lakes



High-quality COI-5P sequences produced a reliable phylogeny confirmed by a heatmap of cophenetic distances

As the COI-5P locus is known to be approximately ~650 bp in length (Folmer et al., 1994), it was imperative to ensure that sequences obtained from BOLD accurately represented the locus of interest, rather than the entire COI gene or the entire mitochondrial genome. Figure 1 displays the impact of filtering on the sequence length and quality. This step effectively selects the most high-quality sequences as representatives for taxa to ensure the downstream alignment and phylogenetic relationships do not introduce systematic errors (Lozano-Fernandez, 2022).

The phylogenetic tree inferred with IQ-TREE2 is consistent with species-level relationships established in the literature (Figure 2). For example, *Coregonus* species form a monophyletic group that is consistent with Backenstose et al. (2024). In contrast, although *Petromyzon* species form a monophyletic group, their position in the tree is not consistent with literature, which supports that lampreys represent an early-diverging vertebrate sister lineage to jawed vertebrates, as shown by Smith et al. (2013). COI-5P is widely used as an animal DNA barcode because it is conserved enough within species but has variation among closely related species. Although it is suitable for species-level identification, it alone is not sufficient for resolving deep divergences (Pentinsaari et al., 2016). A heatmap created from the cophenetic distances inferred from the tree visually confirms distinct groups corresponding to major taxonomic groups (Figure 5).

Fish communities across Great Lakes show shared composition, evolutionary diversity, and somewhat overdispersed phylogenetic structure

Fish communities across the Great Lakes show notable species overlap, with many species present in multiple lakes (Figure 3). This pattern reflects historical connectivity and ecosystem similarity, which is consistent with community composition of other freshwater systems that are influenced by both biogeography and habitat similarity (Lin et al., 2023).

Phylogenetic diversity was positively associated with species richness (Figure 4). This relationship suggests that community assembly processes are not dominated by a single lineage but instead draw from multiple clades, enhancing evolutionary representation.

Nearest Taxon Index values were negative for most lakes, indicating phylogenetic overdispersion, except in lake Ontario, which displays slight clustering (Figure 6). However, these results were non-significant ($NTI > |1.96|$). This pattern implies that coexisting species tend to be somewhat distantly related, consistent with the competitive exclusion hypothesis.

Future work could use multiple loci to recover deeper evolutionary relationships and analyze abiotic factors

Although the results of this analyses were statistically non-significant, the trends displayed in this analysis are consistent with the general hypotheses, suggesting non-random assembly of freshwater fish. Interestingly, most Great Lakes fish assemblages display slight phylogenetic overdispersion, supporting the competitive exclusion hypotheses. However, the lake Ontario community displayed slight phylogenetic clustering, supporting the environmental filtering hypothesis.

A noteworthy limitation for this study was the use of a single mitochondrial locus, COI-5P for phylogenetic inference. Although it was sufficient for identifying species-level relationships and phylogenetic groups, it failed to recover well-established deeper divergences.

Future work could expand on the work done here by:

1. Reconstructing phylogenetic relationships with multiple nuclear and mitochondrial loci with codon-awareness as described by Lozano-Fernandez (2022) and Pentinsaari et al. (2016).
2. Exploring differences in abiotic factors influencing community assembly in the Great Lakes, particularly lake Ontario, which has high human occupation, colder temperatures, and prominent fluctuations in nutrient availability (Centre for Great Lakes Literacy, 2025; Ontario Biodiversity Council, 2024).

Reflection

By completing this assignment and the course overall, I have learned valuable research project management skills and tools for conducting biological analyses. Working through this project end-to-end gave me experience in developing research questions, reviewing existing literature, learning and applying multiple software tools and R packages, generating figures, and interpreting results within a broader scientific context.

The first assignment helped me understand the general layout and workflow for ecological analysis, while the current one pushed me to think critically about how to visualize and communicate findings effectively. I also learned how to be resourceful, adapting and integrating code from multiple sources to meet the needs of my analysis.

Moving forward into my BINF*6999 project and future career, I'll carry several lessons with me:

1. Planning is essential: research design and literature review are fundamental, but creativity is equally important at this stage.
2. Visualizations are powerful tools: they can illustrate processes, validate steps, and clearly communicate results.
3. Versatility matters: there are many tools and approaches for solving a problem, and having technical fluency across many tools makes for a stronger, more adaptable researcher.

Acknowledgements

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I would also like to thank Liona Vu for serving as a second pair of eyes on my code and for testing the full workflow to ensure all analyses ran smoothly and reproducibly.

The following resources were instrumental for building my analysis:

- <https://pedrohbraga.github.io/CommunityPhylogenetics-Workshop/CommunityPhylogenetics-Workshop.html>
- <https://yulab-smu.top/treedata-book/chapter4.html>
- <https://www.molularecologist.com/2017/02/08/phylogenetic-trees-in-r-using-ggtree/>
- <https://bookdown.org/ndphillips/YaRrr/arranging-plots-with-parmfrow-and-layout.html>
- <https://adiradaniel.netlify.app/post/ggmultipane/>
- <https://atrium.lib.uoguelph.ca/server/api/core/bitstreams/2cacda48-793a-4dfa-804d-814b7d724900/content>
- <https://iqtree.github.io/doc/Quickstart>

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